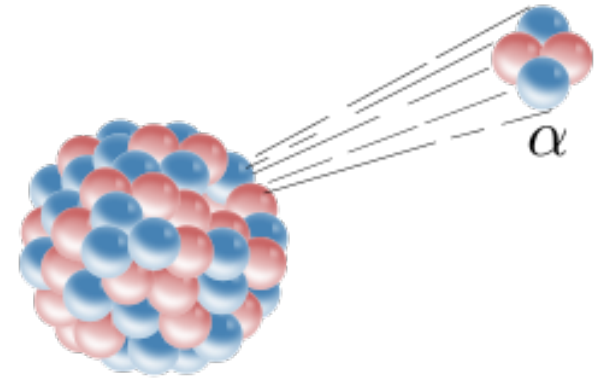


Lecture Unit 3.2

The exponential decay equation

Radioactive decay

- Atoms comprise a central nucleus surrounded by a cloud of electrons
- Some nuclei are radioactive and can spontaneously change into another nucleus by emitting radiation
- The probability of any given nucleus decaying is constant
- The number of decays is proportional to the number of nuclei present



https://en.wikipedia.org/wiki/Alpha_decay

Exponential decay

- When a quantity decreases at a rate which is proportional to the quantity itself it undergoes exponential decay

- The exponential decay of a quantity $N(t)$ is described by the equation

$$\frac{dN}{dt} = -\lambda N \quad \text{where } \lambda \text{ is a positive constant}$$

- Examples include radioactive decay, rates of chemical reactions, heat transfer and many more

Radioactive decay rate

- Number of nuclei as a function of time (N_t) is given by the differential equation

$$\frac{dN_t}{dt} = -\lambda N_t$$

- The time taken for half the nuclei to decay is known as the half-life ($t_{1/2}$)

- The constant λ is related to the half-life by $\lambda = \frac{\ln(2)}{t_{1/2}}$

Numerical approximation

- Approximate the derivative using small but finite differences

$$\frac{dN_t}{dt} \approx \frac{\Delta N_t}{\Delta t}$$

- The change in the number of nuclei over a time interval Δt is given by

$$\Delta N_t \approx \frac{dN_t}{dt} \Delta t$$

$$\text{but } \frac{dN_t}{dt} = -\lambda N_t \text{ so } \Delta N_t \approx -\lambda N_t \Delta t$$

- We can calculate the change in the number of nuclei ΔN_t over a small but finite time interval Δt if we know N_t and λ

Computational solution

- Work out how N_t varies over time by calculating a series of time steps:
 1. Start with N_0 nuclei. Initialise $N_t = N_0$
 2. Calculate rate of change at $t = 0$ using: $\frac{dN_t}{dt} = -\lambda N_t$
 3. Calculate change in number of nuclei (ΔN_t) in a time interval Δt assuming $\frac{dN_t}{dt}$ remains constant over Δt
 4. Update N_t to new value $N_{t+\Delta t}$
 5. Iterate by returning to step 2 until the required end time reached

exp_decay.c

```
#include <stdio.h>
#include <math.h>

float const halfLife=60.0;
float const deltaT=10.0;
int const N0=1e6;
int const maxTimeSteps=12;

int main(){

    float dNdt; /* Rate of change of N*/
    float dN; /* Number of decays in interval deltaT*/

    /* Calculate lambda from half life*/
    float lambda=log(2)/halfLife;
    /* 1. Initialise number of nuclei and time */
    int N=N0;
    float t=0;

    /* Write out initial value of N */
    printf("%f, %d\n", t, N);

    for (int i=0; i<maxTimeSteps; i++){

        /* 2. Calculate rate of change of N */
        dNdt = -1.0 * lambda * N;
        /* 3. Calculate number of decays */
        dN = dNdt * deltaT;
        /* 4. Update number of nuclei and time*/
        N += dN;
        t += deltaT;

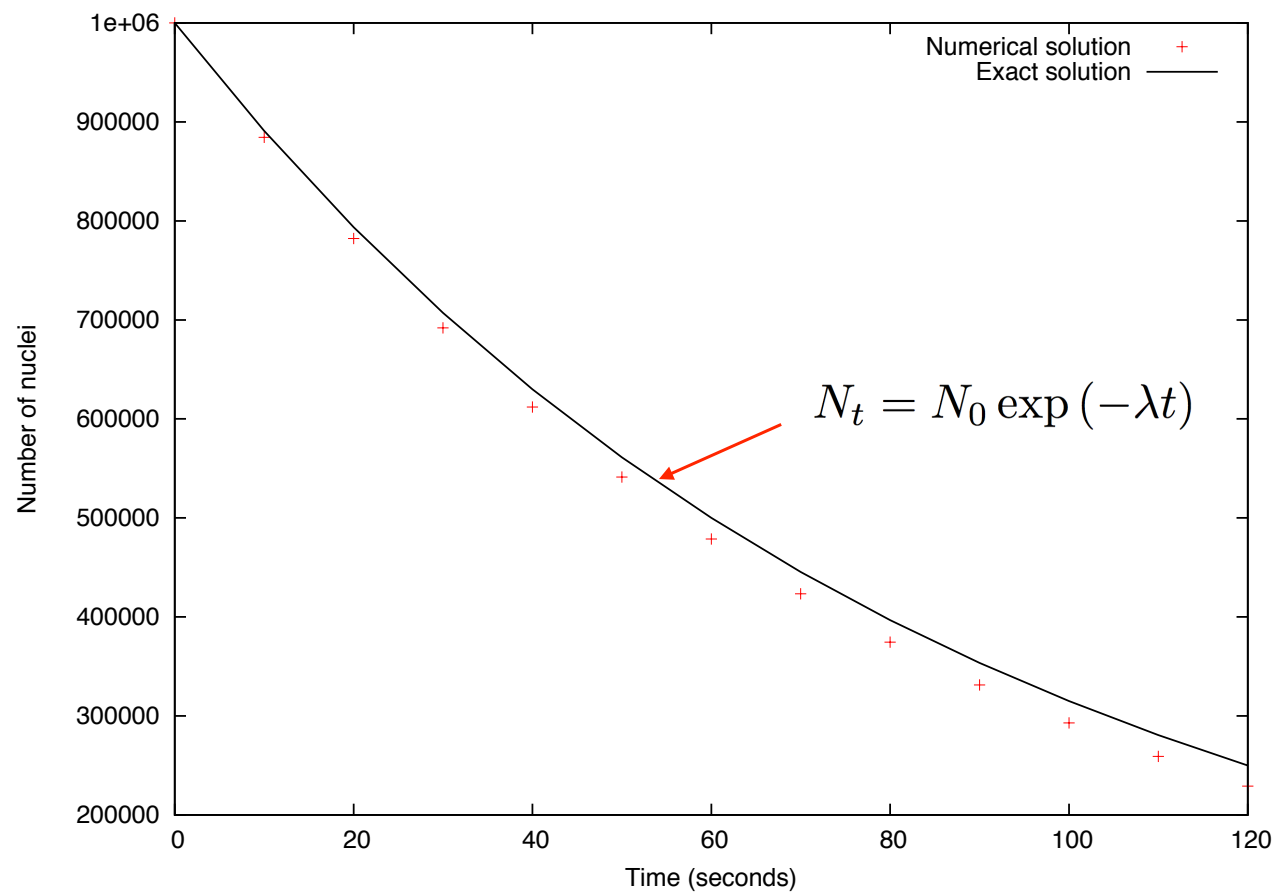
        /* Write out N at current time step */
        printf("%f, %d\n", t, N);
    }

    return 0;
}
```

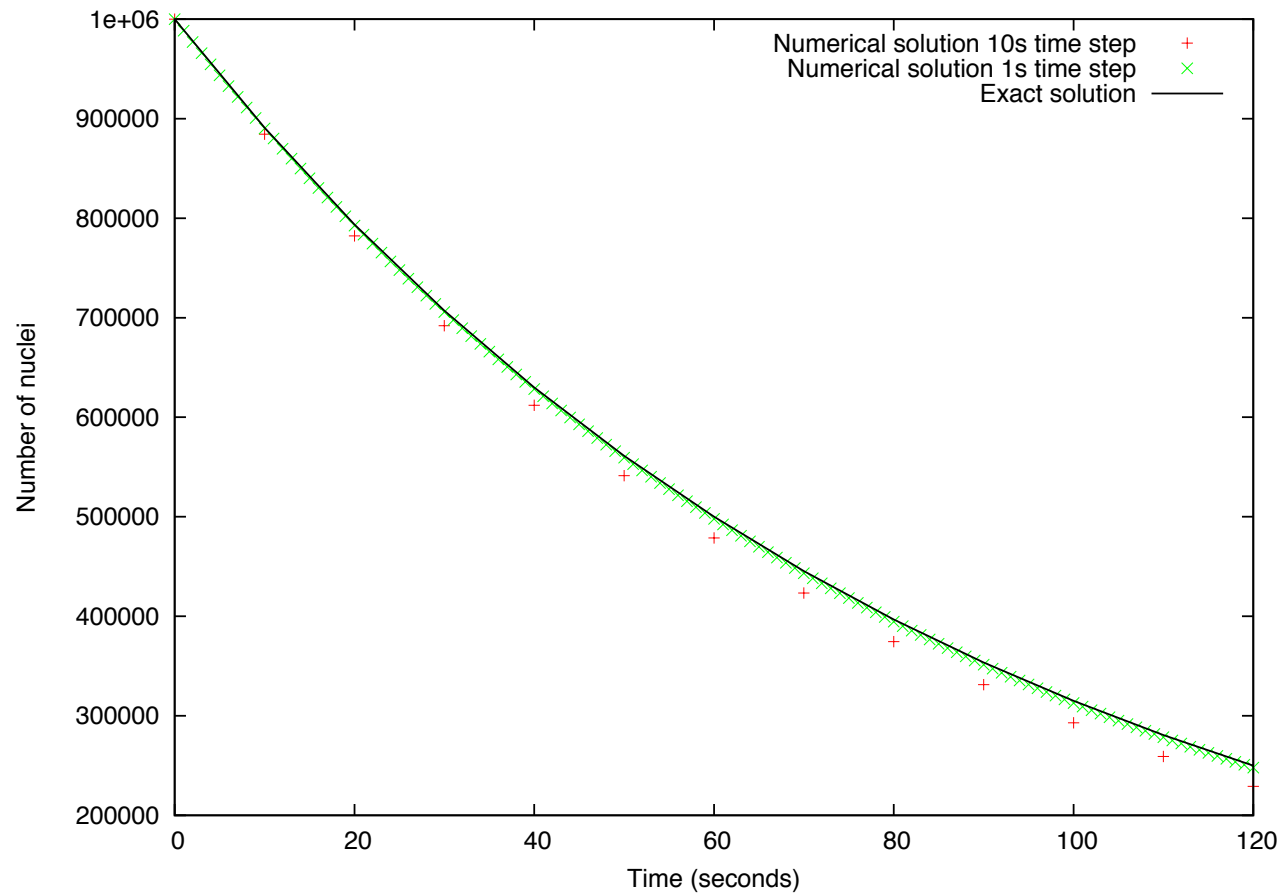
$\log$ is natural logarithm

$$\frac{dN_t}{dt} = -\lambda N_t$$

Results



Results with smaller time-step



Unit Summary

- Exponential decay occurs when a quantity decreases at a rate which is proportional to the quantity itself - radioactive decay is an example
- Radioactive decay is described by a differential equation which gives rate of change of number of nuclei (N_t)
- The equation does not give us a value of N_t directly but instead tells us how N_t changes over time
- We can calculate a numerical solution by specifying an **initial value** and stepping forward in time
- Larger time steps mean less computation but a less accurate result -there is a trade-off